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Plasmonic Porous Metal Nanocomposite

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[MATE6301] Nanomaterials

School of Mechanical and Mining Engineering

The University of Queensland

Plasmonic Porous Metal Nanocomposite

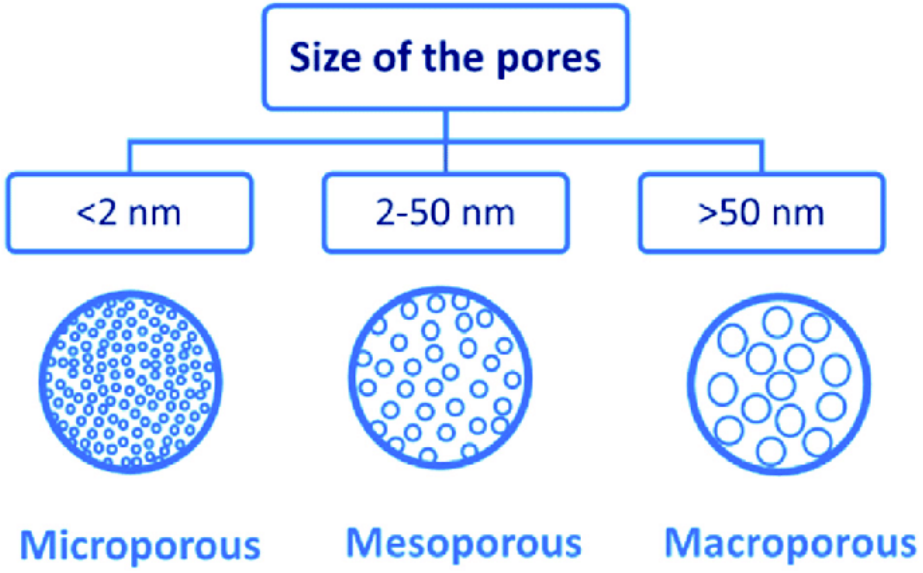


Learning outcomes

- ❑ By the end of this lecture, students should be able to:
- ❑ Explain how porosity, size, shape and composition influence plasmonic behaviour.
- ❑ Explain the origin of SERS enhancement and electromagnetic hotspots.
- ❑ Relate nanostructure design to sensor sensitivity and reproducibility.
- ❑ Compare different SERS sensor architectures.
- ❑ Design a simple plasmonic sensor for an engineering or biomedical application.

ACS Nano 2021, 15(4), 6038–6060

Nanocomposite and porous nanocomposite



The nanocomposite material, which is made of non-metallic, metallic and polymeric materials through a certain method, offers extra benefits!

- High surface-to-volume ratio, which results in small nanofiller sizes and tight filler particle spacing
- Superior mechanical properties (such as high plasticity without sacrificing stability, scratch resistance) and
- Improved optical characteristics.

Materials Today Bio, 2021, *10*, 100100.

Why engineer porosity at the nanoscale?

Porous nanocomposites consist of a regular organic or inorganic bulk phase in which a porous structure is present.

- Nanoporous materials exhibit pore diameters that are most appropriately quantified using units of nanometers.
- The diameter of pores in nanoporous materials is thus typically 100 nanometers or smaller.
- Pores may be open or closed, and pore connectivity and void fraction vary considerably, as with other porous materials.

Structural parameter	**Engineering consequence**
➢ Pore size	✓ Mass transport
➢ Porosity	✓ Surface area
➢ Pore connectivity	✓ Diffusion/accessibility
➢ Ligament size	✓ Mechanical + optical properties
➢ Surface roughness	✓ Electromagnetic hotspots
➢ Metal composition	✓ LSPR wavelength
➢ Structural uniformity	✓ Sensor reproducibility

Materials Today Bio, 2021, *10*, 100100.

Raman Spectroscopy

Raman Spectroscopy provides detailed information about chemical structure, phase and polymorphy, crystallinity and molecular interactions.

- It is based upon the interaction of light with the chemical bonds within a material. Whereby a molecule scatters incident light from a high-intensity laser light source.
- Most of the scattered light is at the same wavelength (or color) as the laser source and does not provide useful information – this is called Rayleigh Scatter.
- However, a small amount of light (typically 0.0000001%) is scattered at different wavelengths (or colors), which depend on the chemical structure of the analyte – this is called Raman Scatter.

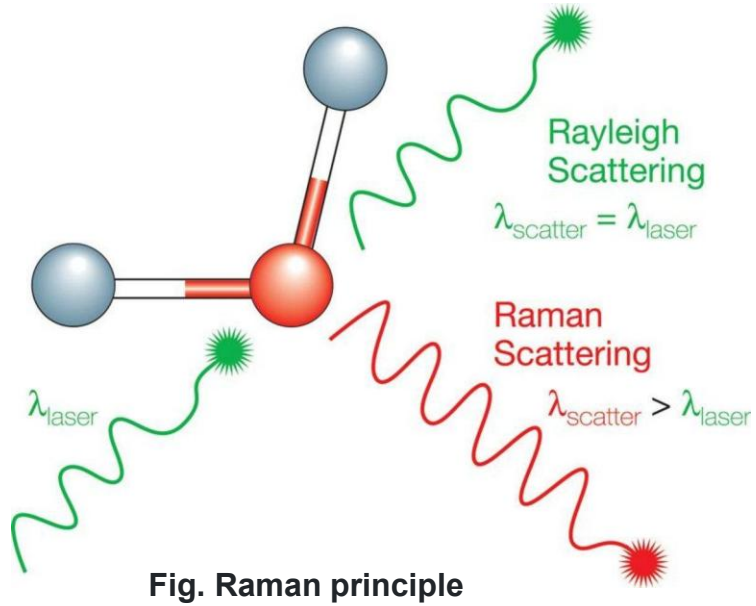


Fig. Raman principle

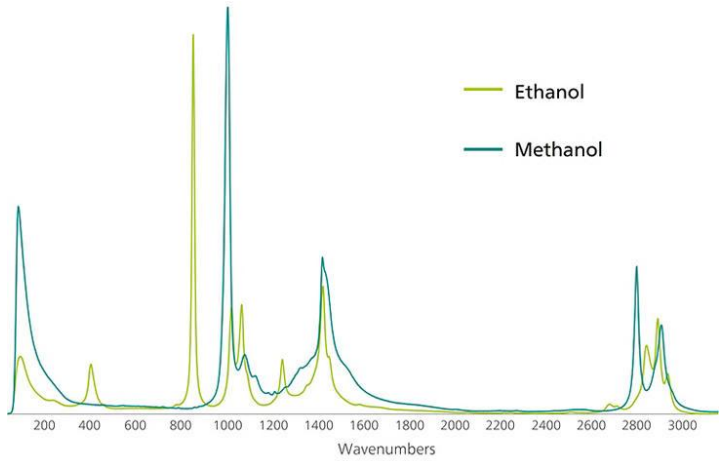


Fig. Raman spectra of ethanol and methanol

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Plasmonic Nanocompisites for Biosensor-Raman versus SERS

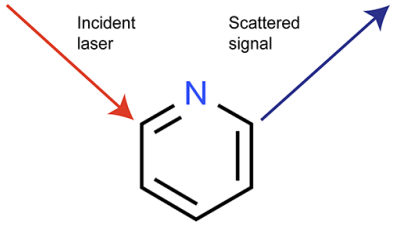
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Raman

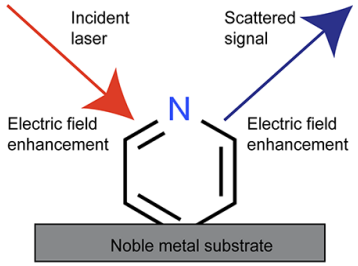
- A laser source is incident directly on a sample.
- The laser light is scattered by the bonds of the analyte, and the inelastically scattered light is collected and processed into a Raman spectrum.
- The non-destructive nature of the technique, the selectivity of Raman bands, and the insensitivity to water make Raman a useful analytical tool for both qualitative and quantitative studies of both organic and inorganic systems.
- Raman spectrum contain more peaks than corresponding SERS spectrum.

SERS

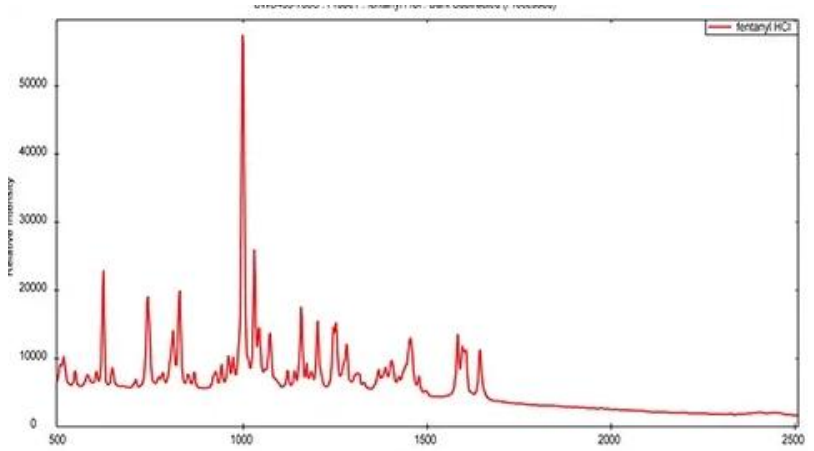
- A incident light falls on the surface of precious metals with a certain amount of roughness.
- SERS enhancement can primarily occur via two mechanisms: electromagnetic enhancement and chemical enhancement.
- In low-concentration sensing where fluorescence overwhelms Raman signal, SERS is an invaluable techniques.
- The SERS bands are noticeably broader than normal Raman bands.



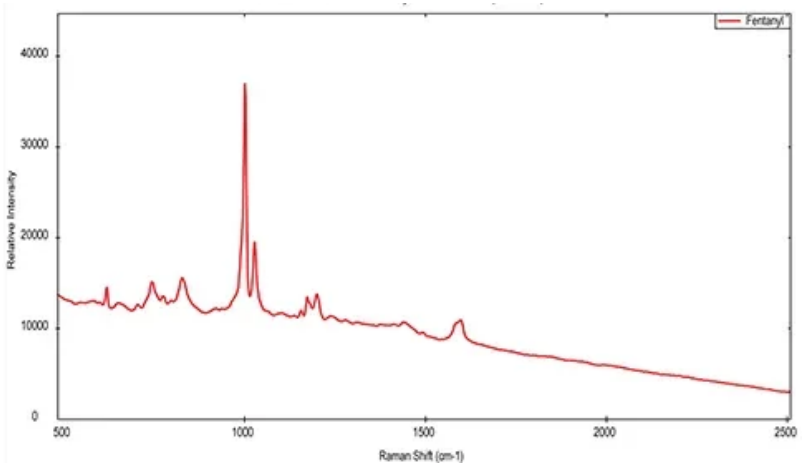
Raman Scattering



Surface-enhanced Raman scattering



Raman

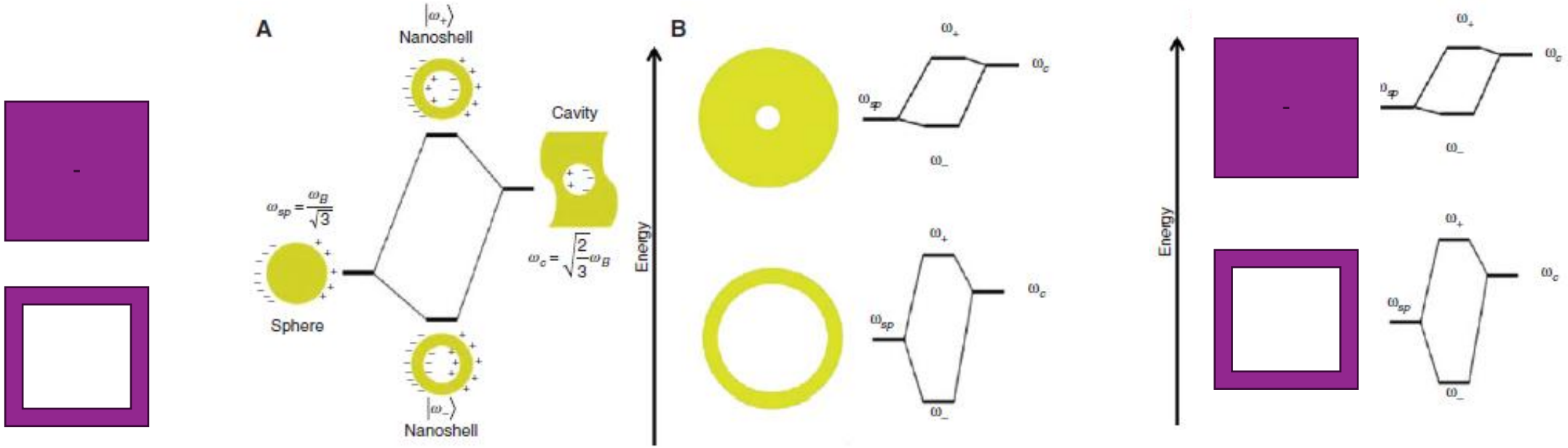


SERS

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Plasmonic properties of composites: Why porous?

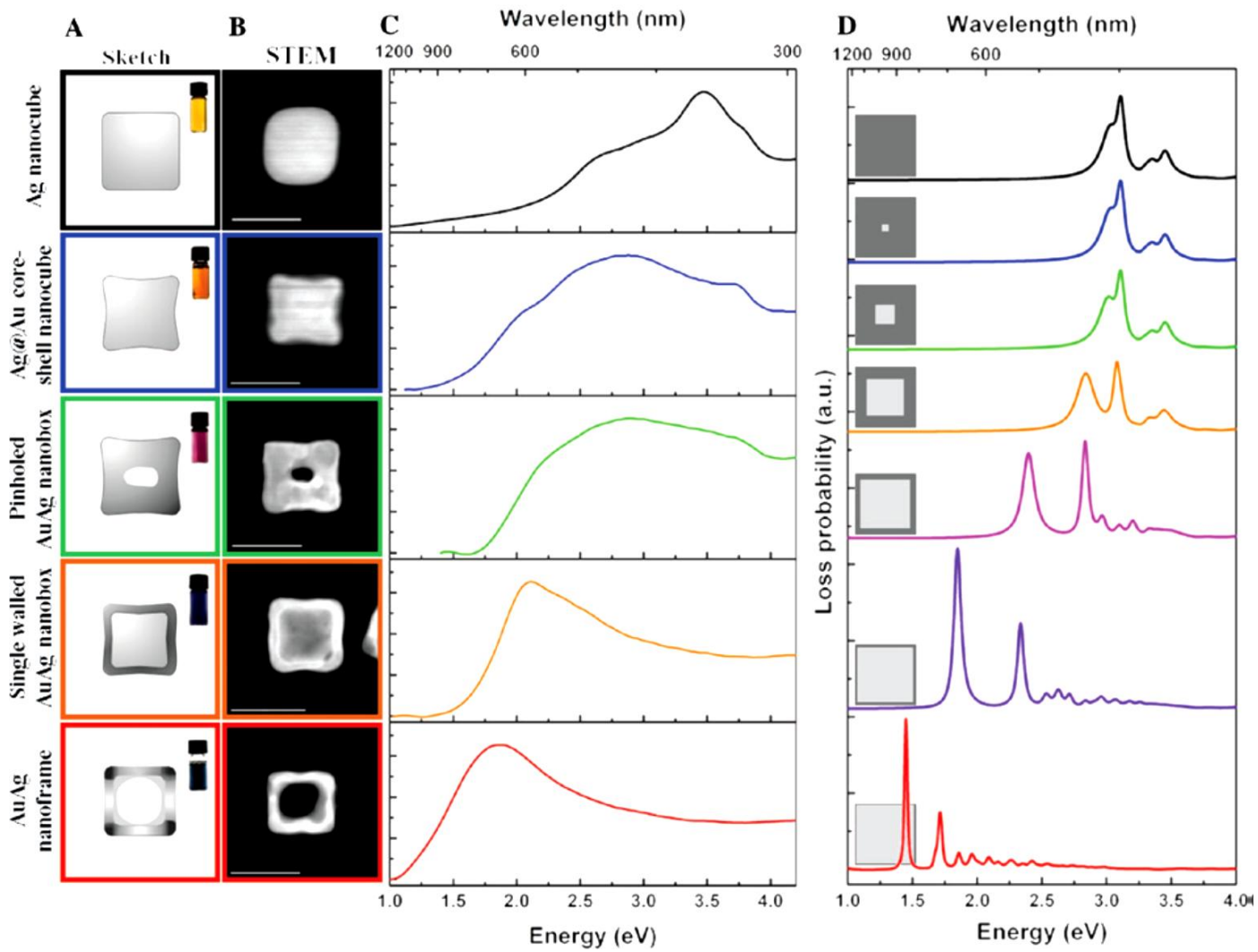
Plasmon Hybridization Mechanism



(A) Schematic energy-level diagram illustrating the plasmon hybridization mechanism in metal nanoshells as a result of the interaction between the sphere (ω_{sp}) and cavity (ω_c) plasmons. Anti-symmetrically coupled, anti-bonding, mode (ω_+) and symmetrically coupled, bonding, mode (ω_-).
(B) Extent of the hybridization for thick (upper) and thin (lower) metallic shells revealing that the hybridization is much more significant for the thin metallic shell.

Nanophotonics 2017; 6(1): 193–213

Porous effect on Plasmonic



Plasmon resonances shift to lower energies with increasing void size. Generally, such an energy shift is the result of two main effects:

(i) compositional effects due to alloying and

(ii) morphological effects due to void formation.

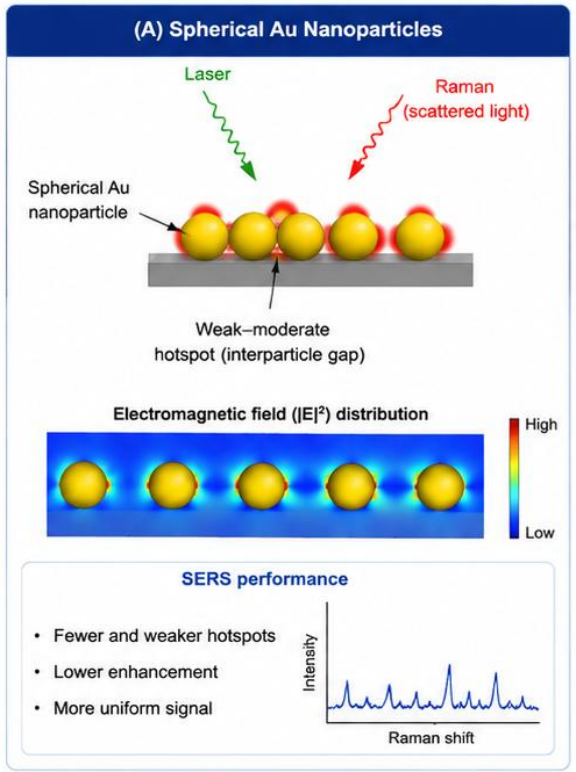
A) Structural sketches and corresponding solution colors from solid Ag nanocubes to single-walled AuAg nanobox and AuAg nanoframe. (B) HAADF STEM images (C) evolution of localized SPRs with structural changes. (D) BEM-simulated EEL spectra revealing the effect of void size for Ag nanostructures.

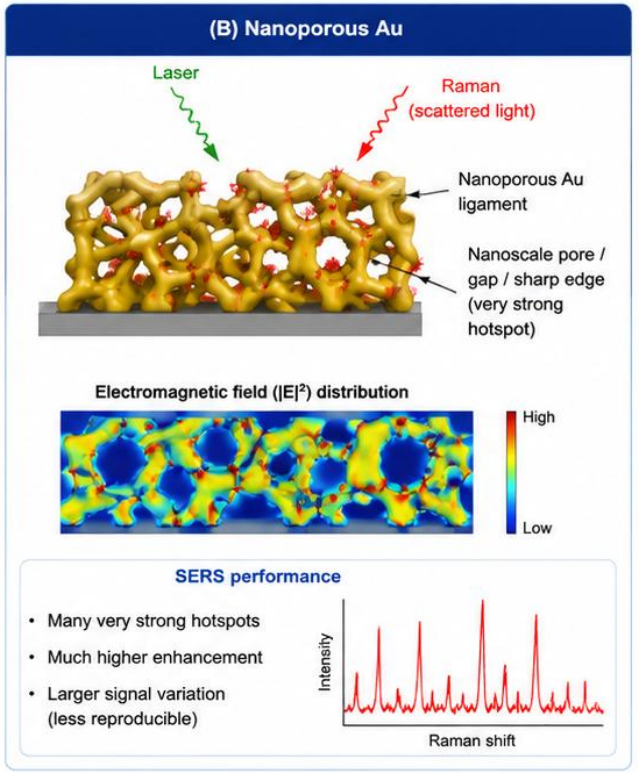
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ACS Photonics 2016;3:770–9.

Porous effect on Plasmonic

- ✓ **Sphere → relatively uniform field → fewer intense hotspots**
- ✓ **Cube / sharp edges → field concentration at corners**
- ✓ **Aggregated particles → interparticle gaps**
- ✓ **Mesoporous metal → many interconnected nanoscale hotspots**





Why nanoporous Au (B) gives stronger SERS

- ✓ Many nanoscale **pores**, gaps and sharp edges
- ✓ Very strong electromagnetic hotspots (**"hot spots"**)
- ✓ Much larger active surface area
- ✓ More analyte adsorption sites
- ✓ More molecules located in hotspots → stronger SERS

ACS Photonics 2016;3:770–9.

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Activity (15 min)

(A) Why can a nanoporous Au structure produce stronger SERS enhancement than spherical Au nanoparticles?

(B) How do pore size, pore connectivity, surface roughness and structural uniformity influence sensing performance?

(C) Design challenge: You need to detect a trace contaminant in water. Which plasmonic structure would you choose—spherical nanoparticles, aggregated nanoparticles or a porous metal substrate? Justify your choice.

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Components of SERS

- Substrate
- Reporter molecule
- Stabilizer molecule
- Conjugating material, and
- Type of fabrication

Substrate

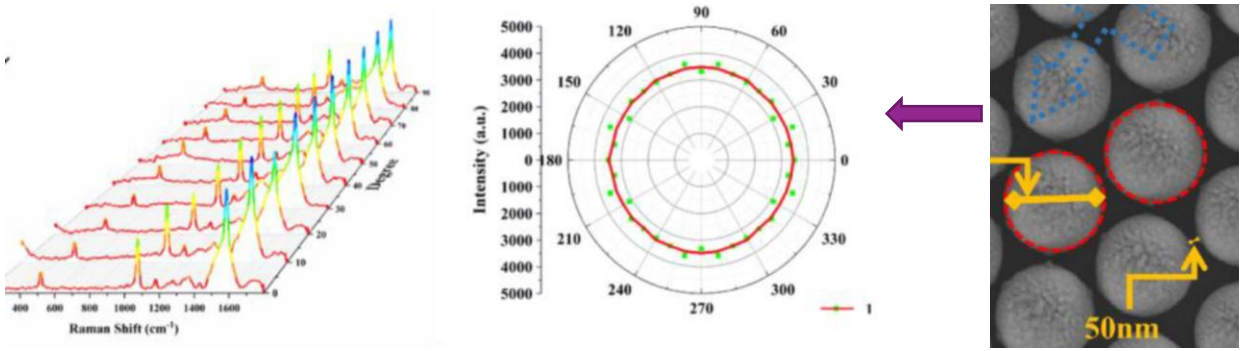
- Substrate is an important component of SERS-based assays.
- It acts as a solid surface that enables the capturing of biomarkers after appropriate functionalization.
- The performance of these substrates strongly depends on the composition of SERS-active materials.
- Ag and Au are widely applied in SERS substrates because of their localized surface plasmon resonance (SPR)
- SPR occurs when the photons of incident light strike the metallic surface at a certain angle of incidence and excite the surface electrons. The exposure of a metal surface to a laser beam scatters conduction electrons into frequency electrons, also known as plasmons, that generate an additional electric field corresponding to the electrons of the laser.
- In nanostructures, these electrons oscillate all over the surface and, thus, are commonly defined as localized surface plasmons, thereby increasing the intensity of the local electromagnetic field, a phenomenon known as localized SPR (LSPR).

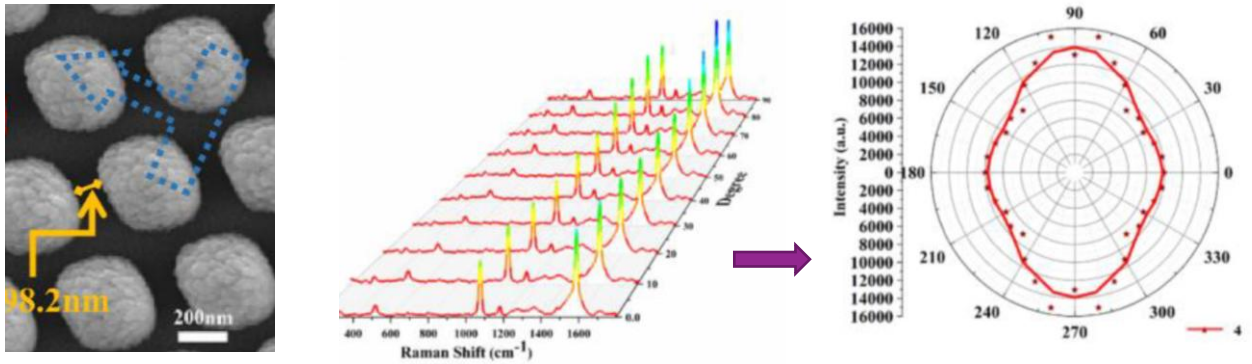
Components of SERS: Substrate

- LSPR depends on the NP size, smaller NPs display poor polarization and result in loss of LSPR properties. The reduced LSPR causes smaller SERS, thereby making these NPs impractical for various SERS-based applications.

- The shape of the NPs also influences the overall SERS efficiency. Spherical NPs have lower efficiency because of the uniform spread of electric field density (hotspots).

- Changing the shape from spherical to cubical NPs improves SERS efficiency, mainly because of the focused non-uniform field close to the sharp ends.

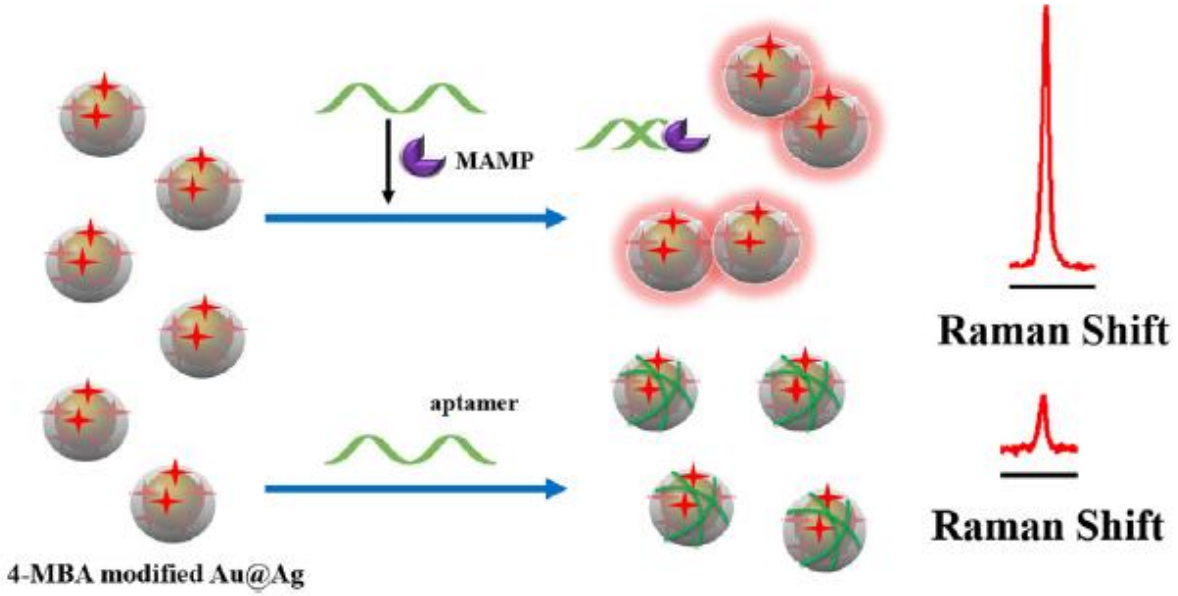




Molecules, 2021, 26, 187

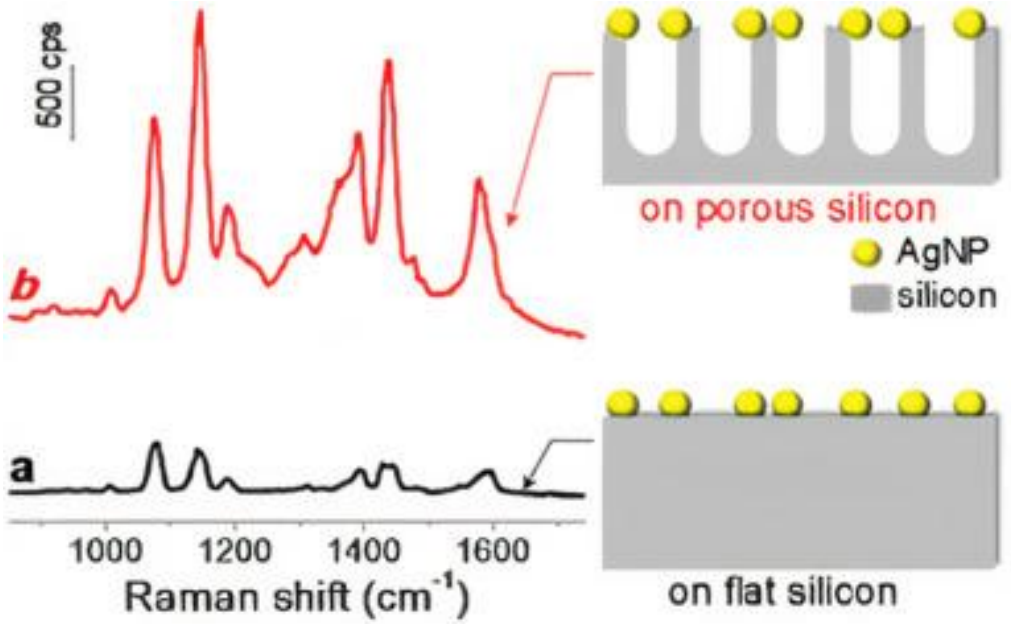
Components of SERS: Aggregation and Porous Substrate

- More hotspots are created with NP aggregation compared with single NPs.



Talanta 190 (2018) 263–268

- Incorporating pores also induces the SERS signal enhancement



Macroporus and Mesoporous Materials, 2021, 111204

Components of SERS: Reporter molecule

The most commonly used reporter molecules are Raman dyes, thiolated reporters, and other small molecules. Usually, Raman active molecules are immobilized on the surface of either AuNPs or AgNPs to form SERS nanotags.

- The functional abilities of Raman reporter molecules (RRMs) include properties such as the formation of self-assembled monolayers (SAMs), elevated Raman signal intensity, and in vivo imaging applications.

- SAM benefits include variability in their molecular orientation and the relative distance of the vibrating group to the metal surface. Thiophenol molecules are particularly relevant for the formation of SAMs. In addition, because of the small number of detectable Raman bands, these thiophenols are preferred when designing multiplex assays.

- Raman dyes [e.g., carboxyfluorescein (FAM), malachite green (MG), rhodamine 6G (R6G), malachite green isothiocyanate (MGITC), methylene blue (MB), and crystal violet (CV)]

- Thiolated reporters [e.g., 5,50-dithiobis(2-nitrobenzoic) (DTNB), 4-nitrothiophenol (NTP4), and mercaptobenzoic acid (MBA), 4-methoxythiophenol. These reporters have the ability to directly conjugate to the surface of metal nanostructures.

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Components of SERS

Stabilizer

To ensure the stability of SERS-encoded particles in biological fluids and prevent the release of the Raman reporter or nonspecific protein adsorption, several polymeric and nonpolymeric media, such as poly(ethylene glycol) (PEG), silica, and graphene, have been used as protective coatings. After the addition of the RRM, the stabilizers are adsorbed onto the surface of the NPs and prevent aggregation, thereby assisting bioconjugation.

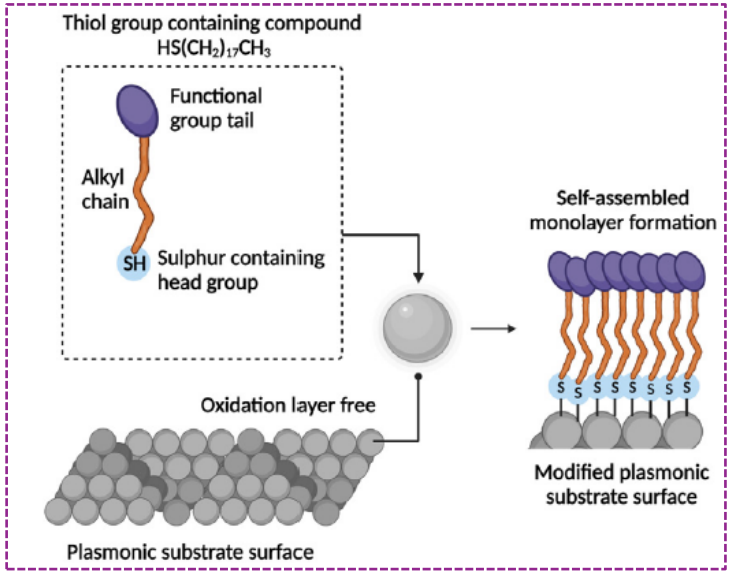
Conjugating Materials

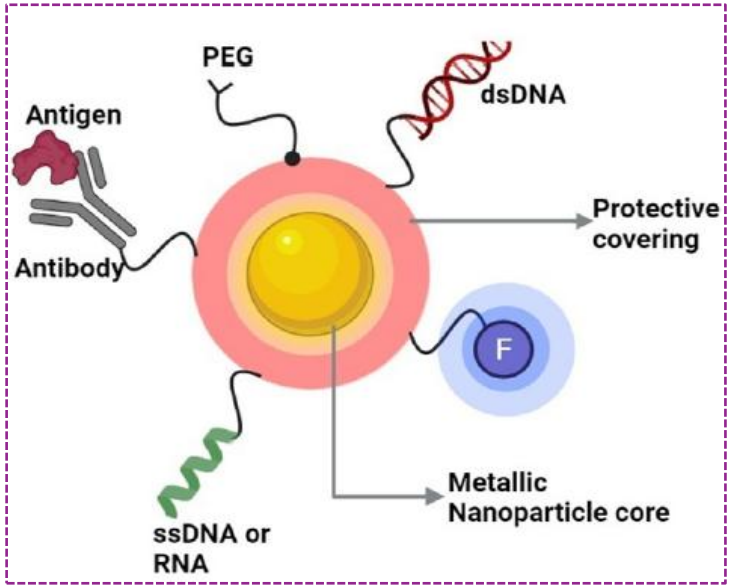
The tag or reporter molecules often have carboxyl groups, thiol groups, or primary amines.

The creation of an amide bond by carbodiimide activation facilitates bio-conjugation of the ligands on the surface of NPs.

The bio-conjugation formed by thiol-group attachment is considered a strong and effective conjugate as it provides better stability.

Other electrostatic interactions and attachment of biotin-streptavidin conjugates to NPs are also used.

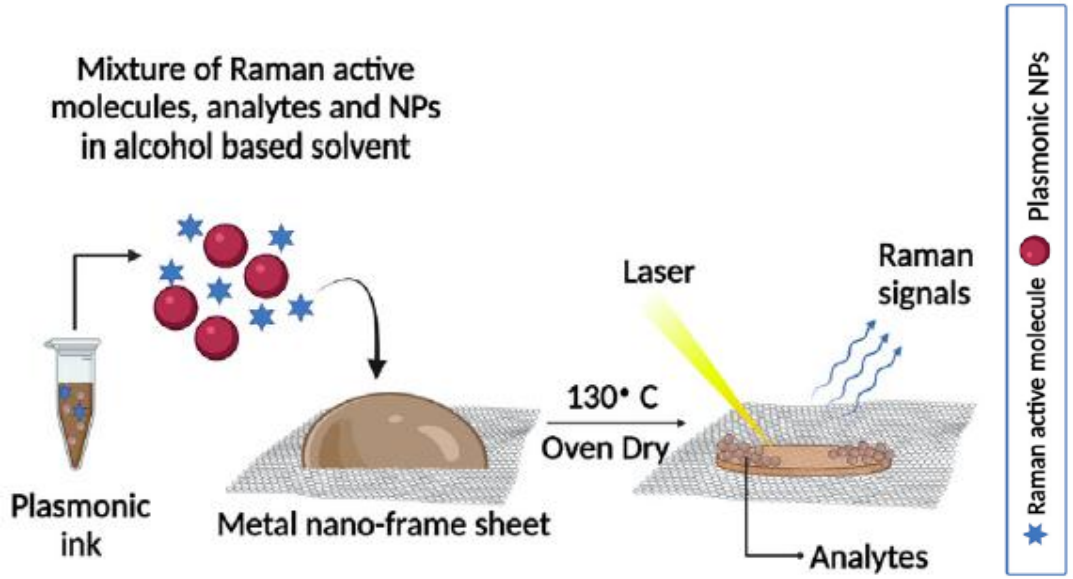




Drug Discovery Today, 2020, 27
(https://doi.org/10.1016/j.drudis.2022.04.016)

Sensors based on unmodified surfaces

The colloid substrates/metal substrate is coated with the standard materials to create a SAM. These SAMs form steady layers around the metal substrate with high coverage and enhance the chemisorption of an analyte to the coated surface.



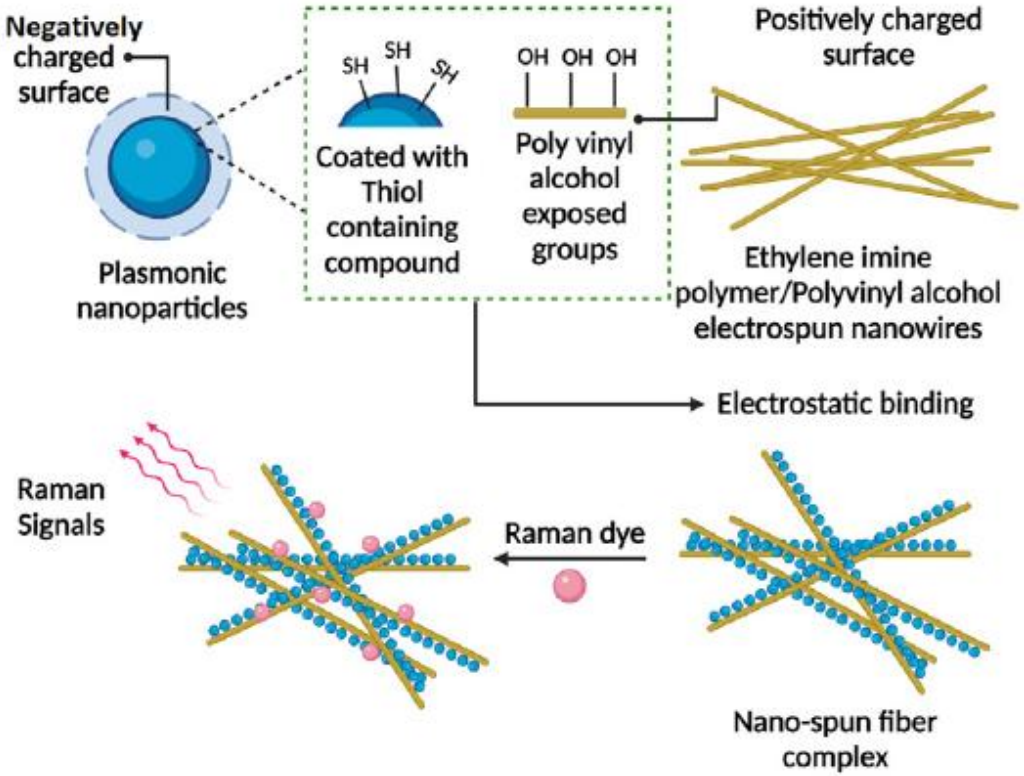
Drug Discovery Today, 2020, 27 (https://doi.org/10.1016/j.drudis.2022.04.016)

Limitation

- Low selectivity- because of imprecise binding and an absence of analyte affinity.
- It is difficult to prepare colloids with uniform sizes and shapes. Thus, the distribution of size and shapes in each batch might differ, which can significantly influence
- the Raman spectrum.
- Uncontrolled and irreproducible colloid aggregation might also make analyte absorption difficult.

Sensors based on modified surfaces

To enhance the selectivity and specificity, functionalized nanostructures with a huge variety of receptor molecules can be used.



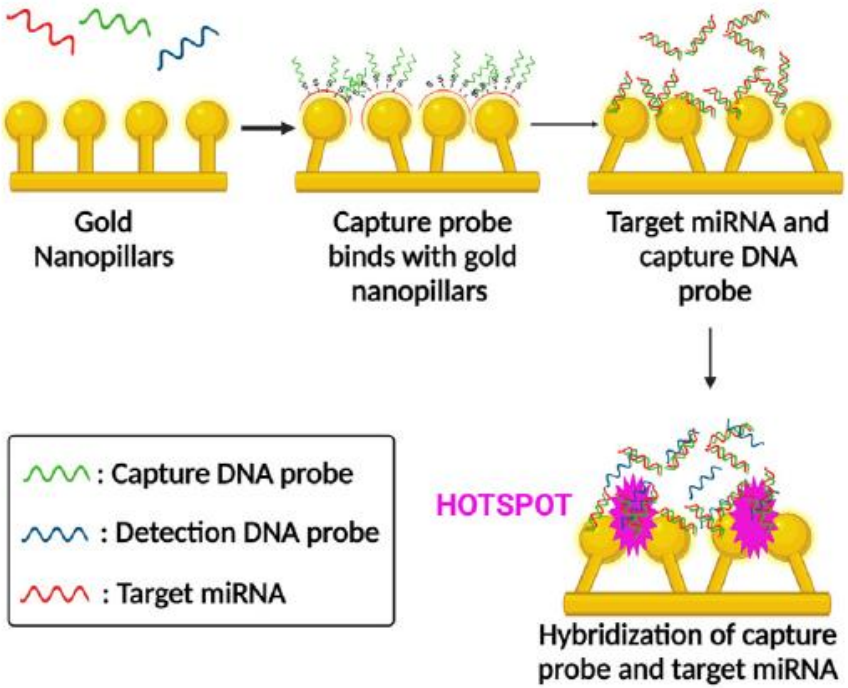
Drug Discovery Today, 2020, 27
(https://doi.org/10.1016/j.drudis.2022.04.016)

Limitation

- Selection of the receptor molecules is limited by the size and the Raman cross-section of the receptor molecules in proportion to the analytes.
- For bulky receptor molecules (e.g., antibodies), the captured molecules might be too distant from the surface to feel the effective increase.

Sensors based on SERS labelling

- SERS label or tags have specific properties, such as photostability, fluorescent potential, and enhanced Raman signals resulting from the in-built hotspots.
- Targeting ligands, including antibodies, aptamers, small molecules, or molecular imprinted polymers, are commonly used in these strategies.
- Enhancement of the Raman signal is directly proportional to the specific binding between an analyte and ligand.



Drug Discovery Today, 2020, 27 (https://doi.org/10.1016/j.drudis.2022.04.016)

Advantage

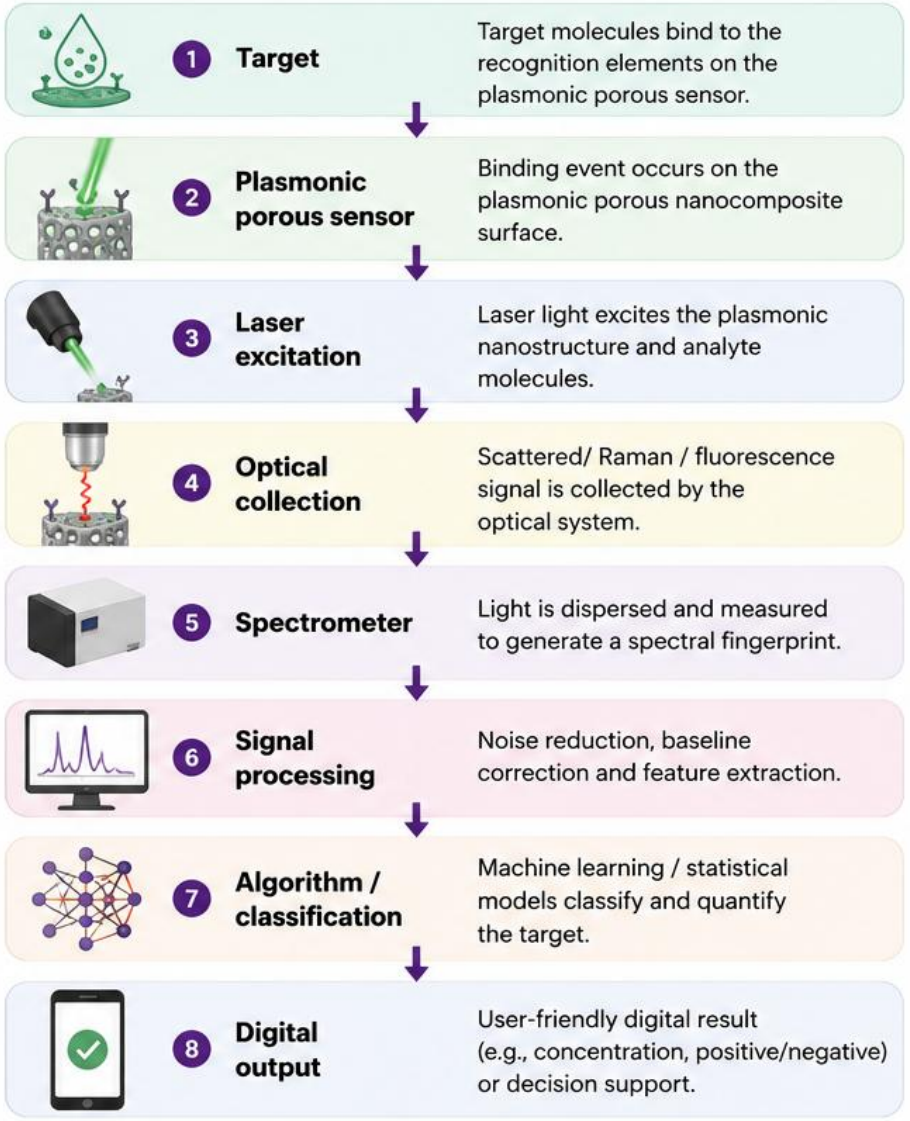
- When the analyte is exposed to the tag, changes such as immobilization of the SERS-tags at specific locations, generation or removal of hotspots, and change in distance between the Raman reporter molecule and the SERS-active substrate are observed.
- The formation of the SERS substrate– ligand complex not only enhances the specificity and sensitivity for trace analysis, but also reduces the chance of false spectra.

SERS-biosensor strategies

Approach	Simplicity	Selectivity	Sensitivity	Complexity
✓ Label-free/unmodified	★★★	★	★★	★
✓ Functionalised surface	★★	★★★	★★	★★
✓ SERS nanotag	★	★★★	★★★	★★★

Which architecture would you choose for a portable sensor and why?

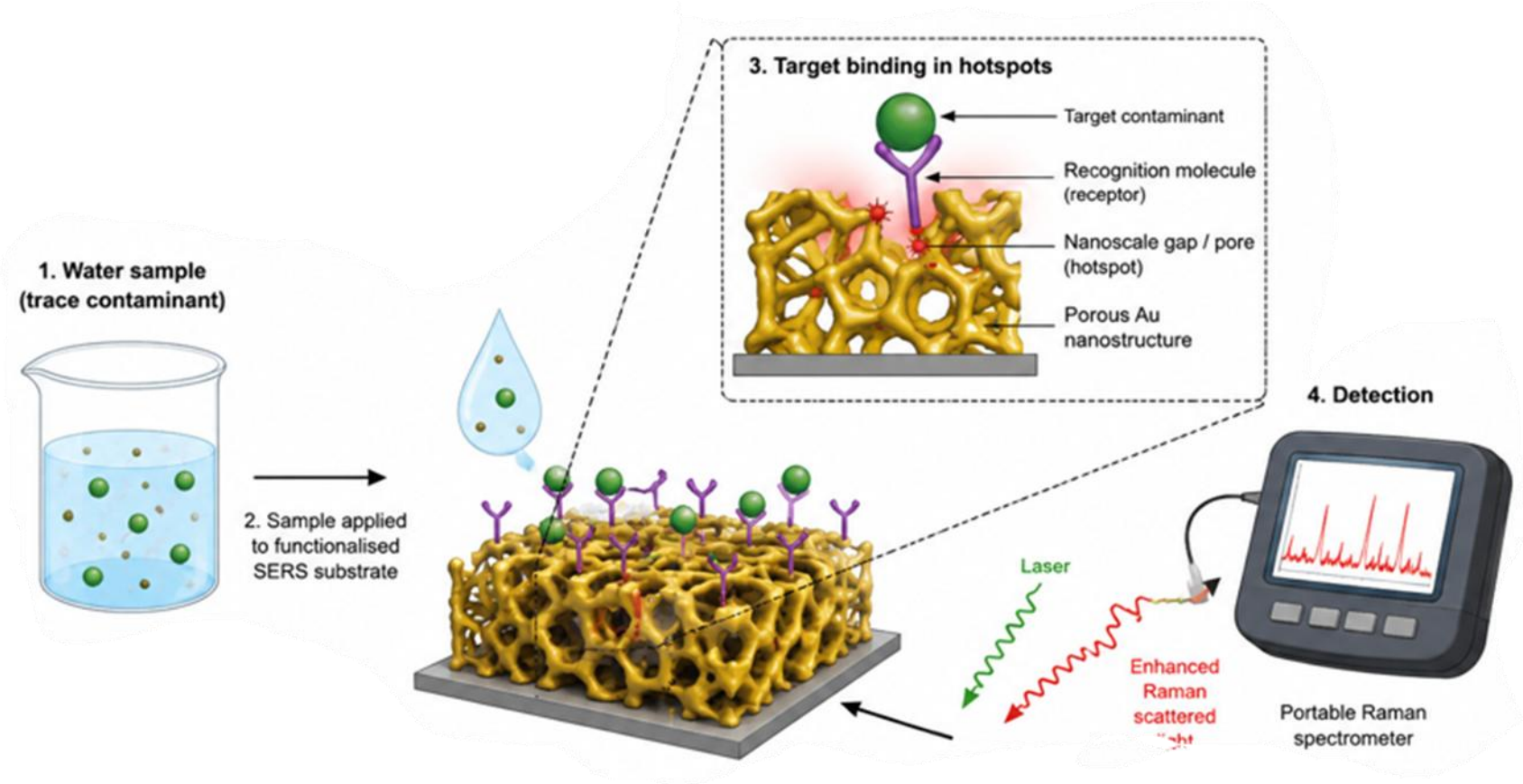
From nanomaterial to smart sensor Target



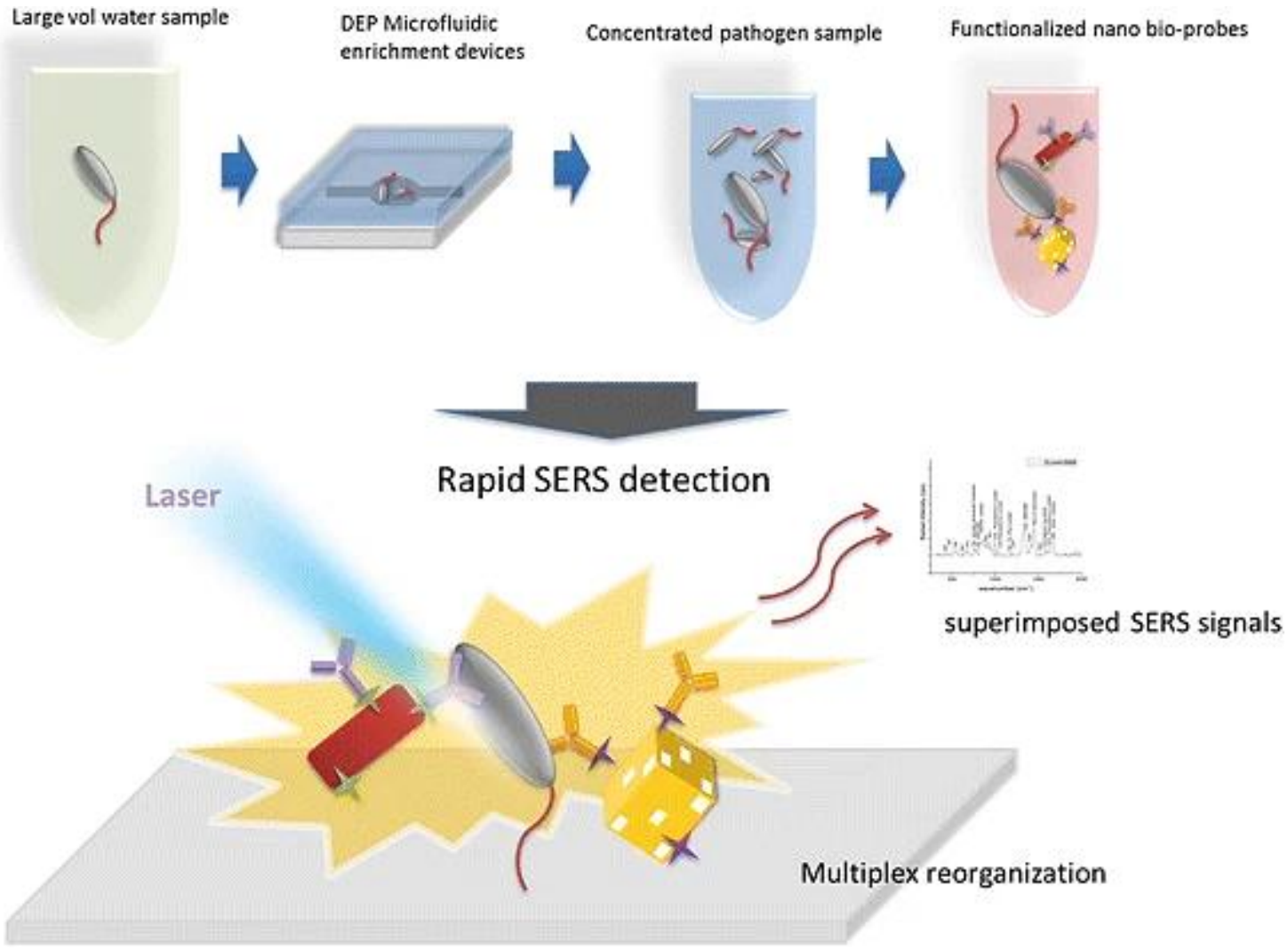
Identify your scope here!

Q. How can you progress this smart sensor?

Porous substrate for water contamination detection

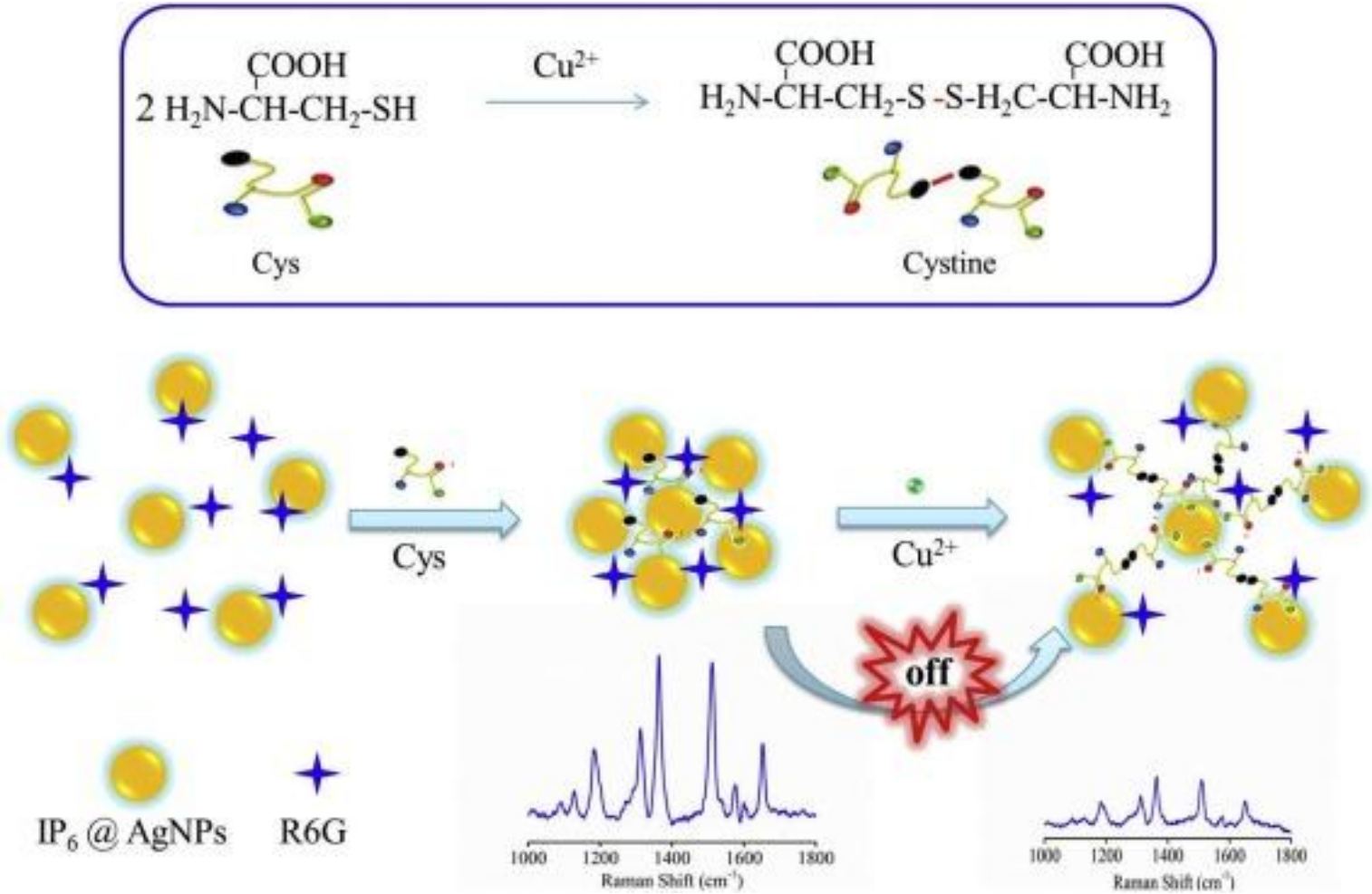


Porous substrate for water borne pathogen detection



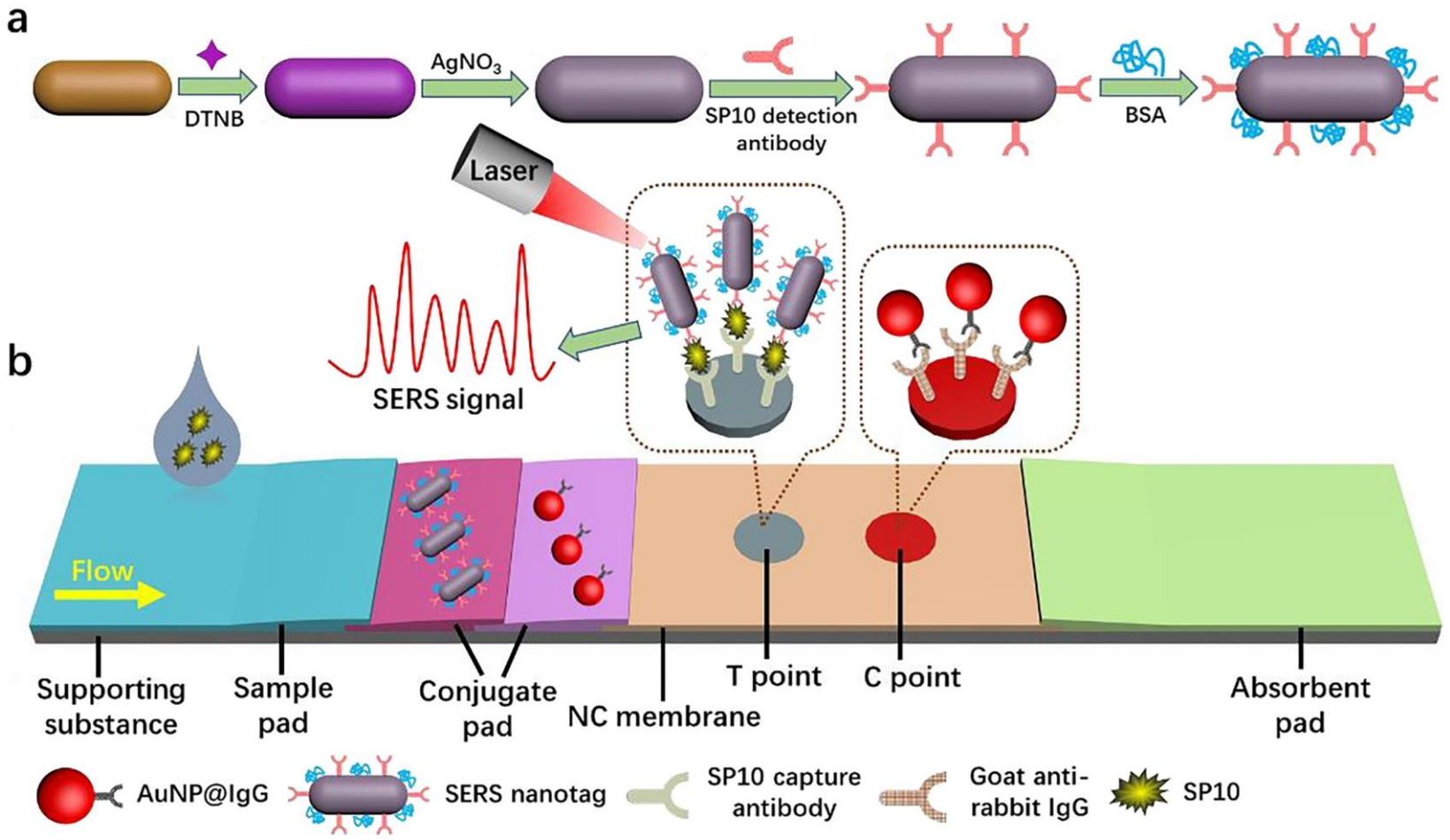
Journal of Biological Engineering, 2017, 11, 9

SERS for heavy metal detection



Sensors and Actuators B: Chemical, 2019, 283, 278-283

SERS-based lateral flow immunoassay strip



Microchemical Journal, **2022**, 175, 107191

CRICOS code 00025B

Activity (10 min)

1. A SERS sensor gives a strong signal but poor reproducibility. What could cause this problem, and how would you improve it?

2. Design challenge: Design a portable SERS sensor for detecting a low-concentration target in a real sample. Explain your choice of plasmonic material, sensing architecture and recognition strategy, and how you would achieve good sensitivity, selectivity and reproducibility.

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Recap and Questions

1. **Structure controls plasmonics: Size • Shape • Porosity • Roughness • Composition**

-determine LSPR electromagnetic hotspots and SERS enhancement.

2. Porosity enhances sensing: High surface area + interconnected pores + nanoscale hotspots

-enable efficient target interaction and strong signal enhancement.

3. Interface controls selectivity: Surface functionalisation + recognition molecules

-bring the target to the plasmonic surface and enable selective detection.

4. Engineering determines real-world performance:

-A successful sensor must balance **Sensitivity • Selectivity • Stability • Reproducibility • Manufacturability**

Nanostructure → Plasmonic Properties → Target Recognition → Signal → Smart Sensor → Real-World Application

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